MONSOON RAINFALL & FLOOD-RISK ANALYSIS FOR MULTAN

Using ERA5 Reanalysis Data

A 10-Year Trend Analysis (2016-2025) with a Focused Case Study of the 2022 Flood Year

Musab Shahid — BES 22-02
Alishba Ehsan — BES 23-34
Mudassir Mujahid — BSES 25-14
Bareera — MPENS 24-02

WHAT DID WE WANT TO KNOW?

① 🌧️ How did Multan's monsoon rainfall vary from 2016 to 2025?
② 📈 Is there evidence of a statistically significant rainfall trend?
③ 🌊 What made the 2022 flood year different?
④ 🔄🌧️ Was 2022 extreme because of one enormous rainfall event — or because of repeated rainfall?

10 YEARS 2016–2025

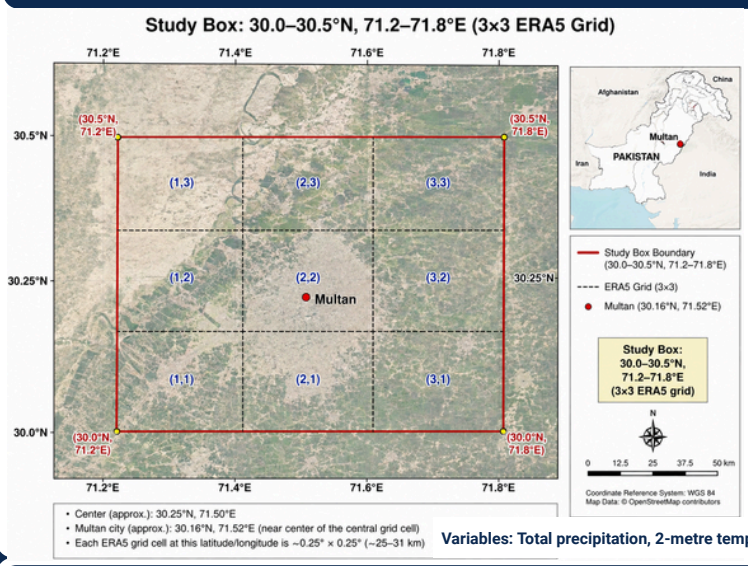
374.2 mm 2022 monsoon rainfall

53.4 mm 2021 — driest monsoon

p = 0.591 No statistically significant rainfall trend

104.3 mm 2020's wettest single day

STUDY AREA & DATA

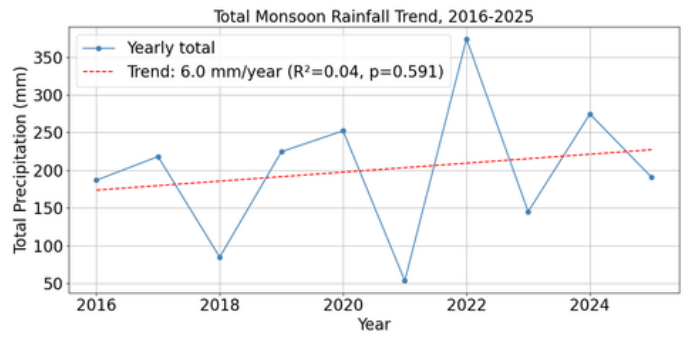


Variables: Total precipitation, 2-metre temperature

METHODOLOGY: FROM RAW DATA TO RESULTS

🛰️ **ERA5 HOURLY DATA**
Precipitation + 2-m temperature
↓
🗺️ **3 × 3 GRID AROUND MULTAN**
Spatial averaging
↓
🔄 **UNIT CONVERSION**
m → mm | K → °C
↓
🕐 **UTC → PKT**
Time conversion
↓
📅 **DAILY AGGREGATION**
Rainfall = SUM | Temperature = MEAN
↓
📊 **INDICES & STATISTICS**
Trend • Heavy Rain • Dry Spells • Extremes

FIGURE 1: 10-YEAR MONSOON RAINFALL TREND (2016-2025)



RESULT: No statistically significant trend
Slope: +5.96 mm/year | R²: 0.038 | p-value: 0.591
Although the fitted line slopes upward, the very low R² and high p-value indicate that the apparent increase cannot be distinguished from random year-to-year variability.

THE 2022 STORY: WHAT MADE THE FLOOD YEAR DIFFERENT?

A. 2022 Compared to the 9-Year Average (2016-2021, 2023-2025)

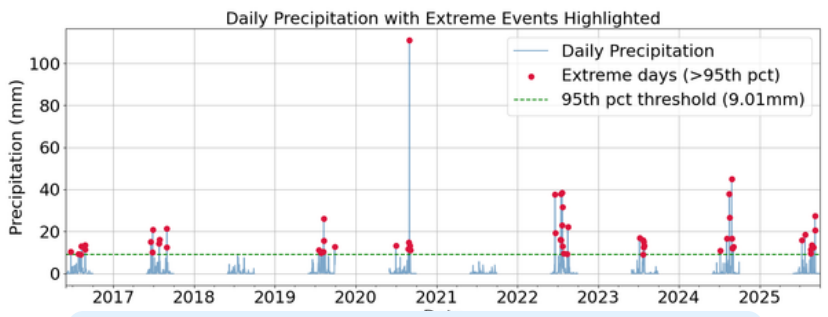
Metric	2022	Average of other 9 years
Total monsoon rainfall	374.2 mm	181.2 mm
Wettest single day	46.7 mm	31.3 mm
Heavy rain days (>10mm)	10	5.1
Longest dry spell	36 days	26.6 days

B. BUT 2022 DID NOT HAVE THE MOST EXTREME DAY

2020 Most Intense Single Day — **104.3 mm** 31 August 2020
VS
2022 Wettest Monsoon Season — **46.7 mm** 21 June 2022

2022's flood-risk signature was not one catastrophic downpour. It was persistent, repeated heavy rainfall across the season.

C. EXTREME RAINFALL DAYS (> 95TH PERCENTILE)



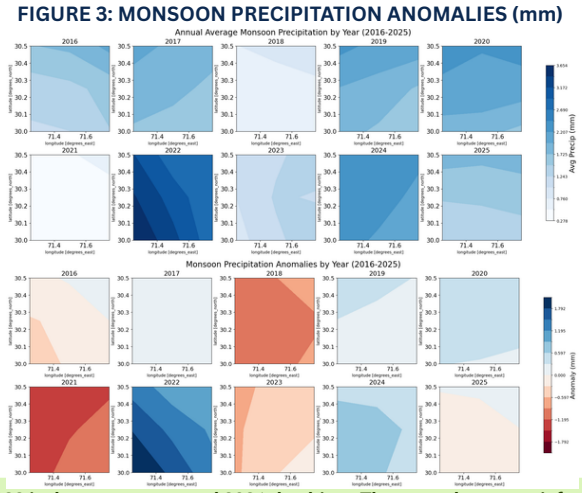
95th-percentile threshold: 9.01 mm/day | 61 days exceeded

Top 5 Most Extreme Events

①	②	③	④	⑤
110.9 mm	44.9 mm	38.4 mm	38.0 mm	37.8 mm
31 Aug 2020	27 Aug 2024	22 Jul 2022	16 Aug 2024	17 Jul 2022

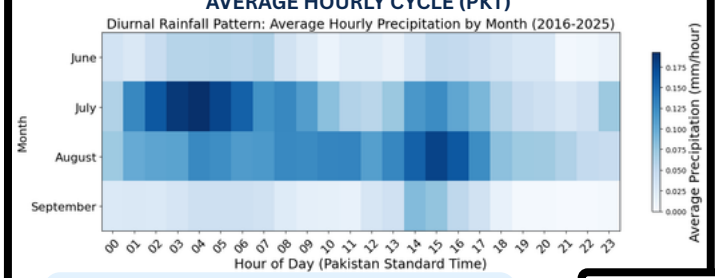
SPATIAL VARIABILITY DURING MONSOON SEASON

FIGURE 2: ANNUAL AVERAGE MONSOON PRECIPITATION (mm)
FIGURE 3: MONSOON PRECIPITATION ANOMALIES (mm)



2022 is the wettest year and 2021 the driest. The anomaly maps reinforce the contrast between these two extremes rather than showing a gradual shift toward wetter conditions. Slight local gradient: slightly wetter toward southwest -+ slightly drier toward northeast.

FIGURE 4: WHEN DOES IT RAIN? AVERAGE HOURLY CYCLE (PKT)



Peak Rainfall Hour (PKT)

June	July	August	September
07:00	04:00	15:00	14:00

OVERALL PEAK: 15:00 (3 pm) PKT

FIGURE 6: MONSOON-SEASON AVERAGE TEMPERATURE (C)

→ 2022: 33.0C (coolest of the decade)
→ 9-year average excluding 2022: 33.9C

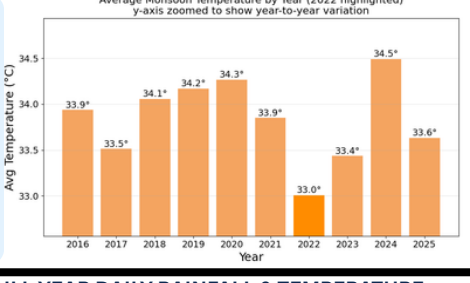
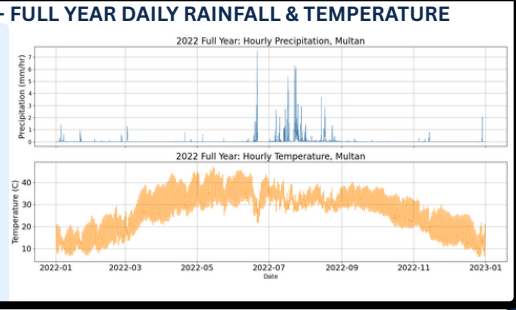


FIGURE 7: 2022 - FULL YEAR DAILY RAINFALL & TEMPERATURE

APRIL-EARLY JUNE
- Highest temperatures >44°C

LATE JUNE ONWARD
- Dense rainfall cluster

MONSOON PERIOD
- Lower temperatures / narrower swings



SOFTWARE & TOOLS
Google Colab, Google Drive, CD6 API, Python (xarray, pandas, NumPy, SciPy, Matplotlib)

DATA SOURCE
🌐 Dataset: ERA5 reanalysis
🛰️ Source: Copernicus Climate Data Store (CDS)
https://cds.climate.copernicus.eu

KEY FINDINGS

🌧️ 2022 was the wettest monsoon year (374.2 mm).
⛈️ 2022 had unusually frequent heavy-rain days (10).
💧 2020 had the most intense single rainfall day (104.3 mm).
📉 No statistically significant rainfall trend for 2016–2025.
📅 Rainfall timing varies substantially by month.
🌡️ Temperature in 2022 was the coolest of the decade during the monsoon season.

WHAT WE CAN CONCLUDE

📊 High year-to-year variability dominates the record.
🌧️🔁 2022's flood-risk signature was persistence and repeated heavy rainfall.
⚠️ Different rainfall signatures can produce different flood-risk outcomes.

WHAT WE CANNOT CONCLUDE

📈🚫 That Multan's long-term climate is definitively becoming wetter.
🌧️🚫 That 2022's flood was caused solely by rainfall in this grid box.
🗺️🚫 How terrain, drainage, river routing, or elevation affected flooding.
🔮🚫 Future climate projections from this analysis.

LIMITATIONS

⏳ 10-year record is relatively short for trend detection.
🗺️ ERA5 grid (~25–31 km) represents area-average, not station-level rainfall.
🌊 Floods depend on many non-climatic factors (terrain, drainage capacity, river behavior).
🚫 No causal attribution or climate projections were made.